Estimation Theory

Nonlinear Kalman Filtering

Extended Kalman Filter (EKF)

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Nonlinear dynamic and measurement model

$$
\begin{aligned}
x_k &= f_{k-1}(x_{k-1}, u_{k-1}, w_{k-1}) \\
y_k &= h_k(x_k, v_k) \\
w_k &\sim (0, Q_k) \\
v_k &\sim (0, R_k)
\end{aligned}
$$

$$
\frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}
$$

Taylor expansion of the dynamic equation around $x_{k-1} = \hat{x}_{k-1}^{+}$ and $w_{k-1} = 0$

$$
\begin{aligned}
x_k &= f_{k-1}(\hat{x}_{k-1}^{+}, u_{k-1}, 0) + \boxed{\left.\frac{\partial f_{k-1}}{\partial x}\right|_{\hat{x}_{k-1}^{+}}}(x_{k-1} - \hat{x}_{k-1}^{+}) + \boxed{\left.\frac{\partial f_{k-1}}{\partial w}\right|_{\hat{x}_{k-1}^{+}}} w_{k-1} \\
&= f_{k-1}(\hat{x}_{k-1}^{+}, u_{k-1}, 0) + \boxed{F_{k-1}}(x_{k-1} - \hat{x}_{k-1}^{+}) + \boxed{L_{k-1}} w_{k-1} \\
&= F_{k-1}x_{k-1} + \left[f_{k-1}(\hat{x}_{k-1}^{+}, u_{k-1}, 0) - F_{k-1}\hat{x}_{k-1}^{+}\right] + L_{k-1}w_{k-1} \\
&= F_{k-1}x_{k-1} + \tilde{u}_{k-1} + \tilde{w}_{k-1}
\end{aligned}
$$

$$
\begin{aligned}
\tilde{u}_k &= f_k(\hat{x}_k^{+}, u_k, 0) - F_k\hat{x}_k^{+} \\
\tilde{w}_k &\sim (0, L_k Q_k L_k^T)
\end{aligned}
$$

Taylor expansion of the measurement equation around $x_k = \hat{x}_k^{-}$ and $v_k = 0$

$$
\begin{aligned}
y_k &= h_k(\hat{x}_k^{-}, 0) + \boxed{\left.\frac{\partial h_k}{\partial x}\right|_{\hat{x}_k^{-}}}(x_k - \hat{x}_k^{-}) + \boxed{\left.\frac{\partial h_k}{\partial v}\right|_{\hat{x}_k^{-}}} v_k \\
&= h_k(\hat{x}_k^{-}, 0) + \boxed{H_k}(x_k - \hat{x}_k^{-}) + \boxed{M_k} v_k \\
&= H_k x_k + \left[h_k(\hat{x}_k^{-}, 0) - H_k\hat{x}_k^{-}\right] + M_k v_k \\
&= H_k x_k + z_k + \tilde{v}_k
\end{aligned}
$$

$$
\begin{aligned}
z_k &= h_k(\hat{x}_k^{-}, 0) - H_k\hat{x}_k^{-} \\
\tilde{v}_k &\sim (0, M_k R_k M_k^T)
\end{aligned}
$$

Linearized model

$$
\begin{aligned}
x_k &= F_{k-1}x_{k-1} + \tilde{u}_{k-1} + \tilde{w}_{k-1} \\
y_k &= H_k x_k + z_k + \tilde{v}_k
\end{aligned}
$$

$$
\begin{aligned}
\tilde{u}_k &= f_k(\hat{x}_k^+, u_k, 0) - F_k \hat{x}_k^+ \\
\tilde{w}_k &\sim (0, L_k Q_k L_k^T)
\end{aligned}
$$

$$
\begin{aligned}
z_k &= h_k(\hat{x}_k^-, 0) - H_k \hat{x}_k^- \\
\tilde{v}_k &\sim (0, M_k R_k M_k^T)
\end{aligned}
$$

$$
\begin{aligned}
F_{k-1} &= \left.\frac{\partial f_{k-1}}{\partial x}\right|_{\hat{x}_{k-1}^+} & H_k &= \left.\frac{\partial h_k}{\partial x}\right|_{\hat{x}_k^-} \\
L_{k-1} &= \left.\frac{\partial f_{k-1}}{\partial w}\right|_{\hat{x}_{k-1}^+} & M_k &= \left.\frac{\partial h_k}{\partial v}\right|_{\hat{x}_k^-}
\end{aligned}
$$

Extended Kalman Filter (EKF)

$$
\begin{aligned}
P_k^- &= F_{k-1} P_{k-1}^+ F_{k-1}^T + L_{k-1} Q_{k-1} L_{k-1}^T \\
K_k &= P_k^- H_k^T (H_k P_k^- H_k^T + M_k R_k M_k^T)^{-1} \\
\hat{x}_k^- &= f_{k-1}(\hat{x}_{k-1}^+, u_{k-1}, 0) \\
z_k &= h_k(\hat{x}_k^-, 0) - H_k \hat{x}_k^- \\
\hat{x}_k^+ &= \hat{x}_k^- + K_k(y_k - H_k \hat{x}_k^- - z_k) \\
&= \hat{x}_k^- + K_k[y_k - h_k(\hat{x}_k^-, 0)] \\
P_k^+ &= (I - K_k H_k) P_k^-
\end{aligned}
$$

Iterated Extended Kalman Filter

Linearization of the measurement equation

$$
\begin{aligned}
h(x_k, v_k) &= h(\hat{x}_k^-, 0) + \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_k^-} (x_k - \hat{x}_k^-) + \left.\frac{\partial h}{\partial v}\right|_{\hat{x}_k^-} v_k \\
&= h(\hat{x}_k^-, 0) + H_k(x_k - \hat{x}_k^-) + M_k v_k
\end{aligned}
$$

Measurement update

$$
\begin{aligned}
K_k &= P_k^- H_k^T (H_k P_k^- H_k^T + M_k R_k M_k^T)^{-1} \\
P_k^+ &= (I - K_k H_k) P_k^- \\
\hat{x}_k^+ &= \hat{x}_k^- + K_k [y_k - h_k(\hat{x}_k^-, 0)]
\end{aligned}
$$

Iterate

better estimate

Initialize

$$
\begin{aligned}
\hat{x}_{k,0}^+ &= \hat{x}_k^+ \\
P_{k,0}^+ &= P_k^+
\end{aligned}
$$

For i = 0, 1, …, N

$$
\begin{aligned}
H_{k,i} &= \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_{k,i}^+} \\
K_{k,i} &= P_k^- H_{k,i}^T (H_{k,i} P_k^- H_{k,i}^T + M_k R_k M_k^T)^{-1} \\
P_{k,i+1}^+ &= (I - K_{k,i} H_{k,i}) P_k^- \\
\hat{x}_{k,i+1}^+ &= \hat{x}_k^- + K_{k,i} [y_k - h_k(\hat{x}_k^-)]
\end{aligned}
$$

Iterated Extended Kalman Filter: An Additional Modification

Measurement equation

$$\begin{aligned} y_k &= h(x_k) \\ &\approx h(\hat{x}_k^-) + H|_{\hat{x}_k^-} (x_k - \hat{x}_k^-) \end{aligned}$$

Measurement equation (Iterated EKF)

$$y_k \approx h(\hat{x}_{k,i}^+) + H|_{\hat{x}_{k,i}^+} (x_k - \hat{x}_{k,i}^+)$$

Residual

$$r_k = y_k - h(\hat{x}_{k,i}^+) - H_{k,i}(\hat{x}_k^- - \hat{x}_{k,i}^+)$$

Update equation (Iterated EKF)

$$\hat{x}_{k,i+1}^+ = \hat{x}_k^- + K_{k,i}[y_k - h(\hat{x}_{k,i}^+) - H_{k,i}(\hat{x}_k^- - \hat{x}_{k,i}^+)]$$

The iterated extended Kalman filter

1. The nonlinear system and measurement equations are given as follows:

$$\begin{aligned} x_k &= f_{k-1}(x_{k-1}, u_{k-1}, w_{k-1}) \\ y_k &= h_k(x_k, v_k) \\ w_k &\sim (0, Q_k) \\ v_k &\sim (0, R_k) \end{aligned} \tag{13.59}$$

2. Initialize the filter as follows.

$$\begin{aligned} \hat{x}_0^+ &= E(x_0) \\ P_0^+ &= E\left[(x_0 - \hat{x}_0)(x_0 - \hat{x}_0)^T\right] \end{aligned} \tag{13.60}$$

3. For $k = 1, 2, \cdots$, do the following.

   (a) Perform the following time-update equations:

$$\begin{aligned} P_k^- &= F_{k-1}P_{k-1}^+F_{k-1}^T + L_{k-1}Q_{k-1}L_{k-1}^T \\ \hat{x}_k^- &= f_{k-1}(\hat{x}_{k-1}^+, u_{k-1}, 0) \end{aligned} \tag{13.61}$$

   where the partial derivative matrices F_{k-1} and L_{k-1} are defined as follows:

$$\begin{aligned} F_{k-1} &= \left.\frac{\partial f_{k-1}}{\partial x}\right|_{\hat{x}_{k-1}^+} \\ L_{k-1} &= \left.\frac{\partial f_{k-1}}{\partial w}\right|_{\hat{x}_{k-1}^+} \end{aligned} \tag{13.62}$$

   Up to this point the iterated EKF is the same as the standard discrete-time EKF.

(b) Perform the measurement update by initializing the iterated EKF estimate to the standard EKF estimate:

$$\begin{aligned} \hat{x}_{k,0}^+ &= \hat{x}_k^- \\ P_{k,0}^+ &= P_k^- \end{aligned} \tag{13.63}$$

For $i = 0, 1, \cdots, N$, evaluate the following equations (where N is the desired number of measurement-update iterations):

$$\begin{aligned} H_{k,i} &= \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_{k,i}^+} \\ M_{k,i} &= \left.\frac{\partial h}{\partial v}\right|_{\hat{x}_{k,i}^+} \\ K_{k,i} &= P_k^- H_{k,i}^T \left(H_{k,i} P_k^- H_{k,i}^T + M_{k,i} R_k M_{k,i}^T\right)^{-1} \\ P_{k,i+1}^+ &= (I - K_{k,i} H_{k,i}) P_k^- \\ \hat{x}_{k,i+1}^+ &= \hat{x}_k^- + K_{k,i}\left[y_k - h(\hat{x}_{k,i}^+) - H_{k,i}(\hat{x}_k^- - \hat{x}_{k,i}^+)\right] \end{aligned} \tag{13.64}$$

(c) The final *a posteriori* state estimate and estimation-error covariance are given as follows:

$$\begin{aligned} \hat{x}_k^+ &= \hat{x}_{k,N+1}^+ \\ P_k^+ &= P_{k,N+1}^+ \end{aligned} \tag{13.65}$$

Higher-Order Approaches

Second-Order Extended Kalman Filter

Hybrid system with continuous-time dynamics and discrete-time measurements

$$\begin{aligned}\dot{x} &= f(x,u,w,t) \\ y_k &= h(x_k,t_k)+v_k \\ w(t) &\sim (0,Q) \\ v_k &\sim (0,R_k)\end{aligned}$$

$$\phi_i = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \leftarrow i\text{th element}$$

Second-order expansion

$$f(x,u,w,t) = f(\hat{x},u_0,w_0,t) + \left.\frac{\partial f}{\partial x}\right|_{\hat{x}}(x-\hat{x}) + \frac{1}{2}\sum_{i=1}^{n}\phi_i (x-\hat{x})^T \left.\frac{\partial^2 f_i}{\partial x^2}\right|_{\hat{x}}(x-\hat{x})$$

$$(x-\hat{x})^T \left.\frac{\partial^2 f_i}{\partial x^2}\right|_{\hat{x}}(x-\hat{x}) = \mathrm{Tr}\left[\left.\frac{\partial^2 f_i}{\partial x^2}\right|_{\hat{x}}(x-\hat{x})(x-\hat{x})^T\right]$$

Unknown

Instead use an expected value

$$(x-\hat{x})^T \left.\frac{\partial^2 f_i}{\partial x^2}\right|_{\hat{x}}(x-\hat{x}) \approx \mathrm{Tr}\left[\left.\frac{\partial^2 f_i}{\partial x^2}\right|_{\hat{x}} P\right]$$

At $x = \hat{x}$,

$$\dot{\hat{x}} = f(\hat{x},u_0,w_0,t) + \frac{1}{2}\sum_{i=1}^{n}\phi_i \mathrm{Tr}\left[\left.\frac{\partial^2 f_i}{\partial x^2}\right|_{\hat{x}} P\right]$$

$$\begin{aligned}\dot{x} &= f(x,u,w,t)\\ y_k &= h(x_k,t_k)+v_k\\ w(t) &\sim (0,Q)\\ v_k &\sim (0,R_k)\end{aligned}$$

Estimation errors

$$\begin{aligned}e_k^- &= x_k - \hat{x}_k^-\\ e_k^+ &= x_k - \hat{x}_k^+\end{aligned}$$

$$\hat{x}_k^+ = \hat{x}_k^- + K_k\left[y_k - h(\hat{x}_k^-, t_k)\right] - \pi_k$$

$$e_k^+ = e_k^- - K_k\left[h(x_k,t_k) - h(\hat{x}_k^-,t_k)\right] - K_k v_k + \pi_k$$

A correction term to make $\hat{x}_k^+$ unbiased

Second-order expansion

$$\begin{aligned}h(x_k,t_k) &= h(\hat{x}_k^-,t_k) + \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_k^-}(x_k-\hat{x}_k^-) + \\ &\quad \frac{1}{2}\sum_{i=1}^{m}\phi_i(x_k-\hat{x}_k^-)^T \left.\frac{\partial^2 h(i)}{\partial x^2}\right|_{\hat{x}_k^-}(x_k-\hat{x}_k^-)\\ &= h(\hat{x}_k^-,t_k) + H_k(x_k-\hat{x}_k^-) + \frac{1}{2}\sum_{i=1}^{m}\phi_i(x_k-\hat{x}_k^-)^T\left.\frac{\partial^2 h_i}{\partial x^2}\right|_{\hat{x}_k^-}(x_k-\hat{x}_k^-)\end{aligned}$$

$$e_k^+ = e_k^- - K_k H_k e_k^- - \frac{1}{2}K_k\sum_{i=1}^{m}\phi_i(e_k^-)^T D_{k,i}e_k^- - K_k v_k + \pi_k \qquad D_{k,i} = \left.\frac{\partial^2 h_i}{\partial x^2}\right|_{\hat{x}_k^-}$$

To make $\mathbb{E}(e_k^+)=0$, we need $\pi_k = \frac{1}{2}K_k\sum_{i=1}^{m}\phi_i \mathrm{Tr}\left[D_{k,i}P_k^-\right]$ assuming $\mathbb{E}(e_k^-)=0$.

$$P_k^+ = E\left[e_k^+(e_k^+)^T\right]$$

New

It can be shown that

$$P_k^+ = (I - K_kH_k)P_k^-(I - K_kH_k)^T + K_k(R_k + \Lambda_k)K_k^T$$

$$\Lambda_k = \frac{1}{4}E\left\{\left[\sum_{i=1}^{m}\phi_i \mathrm{Tr}[D_{k,i}(e_k^-(e_k^-)^T - P_k^-)]\right]\left[\cdots\right]^T\right\}$$

$$\Lambda_k(i,j) = \frac{1}{2}\mathrm{Tr}(D_{k,i}P_k^-D_{k,j}P_k^-)$$

Cost function

$$\begin{aligned} J_k &= E[(e_k^+)^T S_k e_k^+] \\ &= \mathrm{Tr}[S_k P_k^+] \end{aligned}$$

Kalman gain which minimizes the cost function

$$K_k = P_k^- H_k^T \left(H_k P_k^- H_k^T + R_k + \Lambda_k\right)^{-1}$$

$$P_k^+ = P_k^- - P_k^- H_k^T \left(H_k P_k^- H_k^T + R_k + \Lambda_k\right)^{-1} H_k P_k^-$$

Example: Falling Body Problem

$$\begin{aligned}\dot{x}_1 &= x_2 + w_1\\ \dot{x}_2 &= \rho_0 \exp(-x_1/k)x_2^2 x_3/2 - g + w_2\\ \dot{x}_3 &= w_3\end{aligned}$$

$$\begin{aligned}y_k &= \sqrt{M^2 + (x_1(t_k) - a)^2} + v_k\\ &= h(x_k) + v_k\end{aligned}$$

$$\begin{aligned}A &= \frac{\partial f}{\partial x}\\ &= \begin{bmatrix} 0 & 1 & 0\\ A_{21} & A_{22} & A_{23}\\ 0 & 0 & 0\end{bmatrix}\\ A_{21} &= -\rho_0 \exp(-x_1/k)x_2^2/2kx_3\\ A_{22} &= \rho_0 \exp(-x_1/k)x_2/x_3\\ A_{23} &= -\rho_0 \exp(-x_1/k)x_2^2/2x_3^2\end{aligned}$$

$$\begin{aligned}L &= \frac{\partial f}{\partial w}\\ &= \begin{bmatrix} 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}\end{aligned}$$

$$\begin{aligned}H &= \frac{\partial h}{\partial x}\\ &= \begin{bmatrix} (x_1 - a)(M^2 + (x_1 - a)^2)^{-1/2} & 0 & 0\end{bmatrix}\\ D_1 &= \frac{\partial^2 h_1}{\partial x^2}\\ &= \begin{bmatrix} h^{-1}(1 - (x_1 - a)^2 h^{-2}) & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}\end{aligned}$$

$$\frac{\partial^2 f_1}{\partial x^2} = \frac{\partial^2 f_3}{\partial x^2} = \begin{bmatrix} 0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$$

$$\frac{\partial^2 f_2}{\partial x^2} = \rho_0 \exp(-x_1/k)\begin{bmatrix} x_2^2 x_3/2k^2 & -x_2 x_3/k & -x_2^2/2k\\ -x_2 x_3/k & x_3 & x_2\\ -x_2^2/2k & x_2 & 0\end{bmatrix}$$

Estimation errors

Filter	Altitude	Velocity	Ballistic Coefficient
First-order EKF	758 feet	518 feet/sec	0.091 $\text{feet}^3/\text{lb}/\text{sec}^2$
Second-order EKF	356	483	0.129
Iterated EKF ($N = 2$)	755	517	0.091
Iterated EKF ($N = 3$)	745	516	0.091
Iterated EKF ($N = 4$)	738	509	0.091
Iterated EKF ($N = 5$)	733	506	0.091
Iterated EKF ($N = 6$)	723	506	0.091

The second-order discrete-time extended Kalman filter

1. The system equations are given as follows:

$$\begin{aligned} x_{k+1} &= f(x_k, u_k, k) + w_k \\ y_k &= h(x_k, k) + v_k \\ w_k &\sim (0, Q_k) \\ v_k &\sim (0, R_k) \end{aligned}$$

2. The estimator is initialized as follows:

$$\begin{aligned} \hat{x}_0^+ &= E(x_0) \\ P_0^+ &= E\left[(x_0 - \hat{x}_0^+)(x_0 - \hat{x}_0^+)^T\right] \end{aligned}$$

3. The time update equations are given as follows:

$$\begin{aligned} \hat{x}_{k+1}^- &= f(\hat{x}_k^+, u_k, k) + \frac{1}{2}\sum_{i=1}^{n} \phi_i \mathrm{Tr}\left[\left.\frac{\partial^2 f_i}{\partial x}\right|_{\hat{x}_k^+} P_k^+\right] \\ P_{k+1}^- &= F P_k^+ F^T + Q_k \end{aligned}$$

$$\begin{aligned} \phi_i &= \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \begin{matrix} \\ \\ \\ \leftarrow i\text{th element} \\ \\ \\ \\ \end{matrix} \\ F &= \left.\frac{\partial f}{\partial x}\right|_{\hat{x}_k^+} \end{aligned}$$

The measurement update equations are given as follows:

$$\begin{aligned} \hat{x}_k^+ &= \hat{x}_k^- + K_k[y_k - h(\hat{x}_k^-, k)] - \pi_k \\ \pi_k &= \frac{1}{2}K_k \sum_{i=1}^{m} \phi_i \mathrm{Tr}\left[D_{k,i} P_k^-\right] \\ D_{k,i} &= \left.\frac{\partial^2 h_i(x_k, k)}{\partial x^2}\right|_{\hat{x}_k^-} \\ K_k &= P_k^- H_k^T \left(H_k P_k^- H_k^T + R_k\right)^{-1} \\ H_k &= \left.\frac{\partial h(x_k, k)}{\partial x}\right|_{\hat{x}_k^-} \\ P_k^+ &= (I - K_k H_k) P_k^- \end{aligned}$$

The Gaussian sum filter

1. The discrete-time n-state system and measurement equations are given as follows:

$$\begin{aligned} x_k &= f_{k-1}(x_{k-1}, u_{k-1}, w_{k-1}) \\ y_k &= h_k(x_k, v_k) \\ w_k &\sim (0, Q_k) \\ v_k &\sim (0, R_k) \end{aligned} \tag{13.102}$$

2. Initialize the filter by approximating the pdf of the initial state as follows:

Mixture of Gaussians

$$\mathrm{pdf}(\hat{x}_0^+) = \sum_{i=1}^{M} a_{0i} N(\hat{x}_{0i}^+, P_{0i}^+) \tag{13.103}$$

The a_{0i} coefficients (which are positive and add up to 1), the $\hat{x}_{0i}^+$ means, and the P_{0i}^+ covariances, are chosen by the user to provide a good approximation to the pdf of the initial state.

3. For $k = 1, 2, \cdots$, do the following.

   (a) The *a priori* state estimate is obtained by first executing the following time-update equations for $i = 1, \cdots, M$:

$$\begin{aligned}
\hat{x}_{ki}^- &= f_{k-1}(\hat{x}_{k-1,i}^+, u_{k-1}, 0) \\
F_{k-1,i} &= \left.\frac{\partial f_{k-1}}{\partial x_{k-1}}\right|_{\hat{x}_{k-1,i}^+} \\
P_{ki}^- &= F_{k-1,i} P_{k-1,i}^+ F_{k-1,i}^T + Q_{k-1} \\
a_{ki} &= a_{k-1,i}
\end{aligned} \tag{13.104}$$

   The pdf of the *a priori* state estimate is obtained by the following sum:

$$\text{pdf}(\hat{x}_k^-) = \sum_{i=1}^{M} a_{ki} N(\hat{x}_{ki}^-, P_{ki}^-) \tag{13.105}$$

   (b) The *a posteriori* state estimate is obtained by first executing the following measurement update equations for $i = 1, \cdots, M$:

$$\begin{aligned}
H_{ki} &= \left.\frac{\partial h_k}{\partial x_k}\right|_{\hat{x}_{ki}^-} \\
K_{ki} &= P_{ki}^- H_{ki}^T (H_{ki} P_{ki}^- H_{ki}^T + R_k)^{-1} \\
P_{ki}^+ &= P_{ki}^- - K_{ki} H_{ki} P_{ki}^- \\
\hat{x}_{ki}^+ &= \hat{x}_{ki}^- + K_{ki} \left[y_k - h_k(\hat{x}_{ki}^-, 0) \right]
\end{aligned} \tag{13.106}$$

$$\text{pdf}(\hat{x}_k^+) = \sum_{i=1}^{M} a_{ki} N(\hat{x}_{ki}^+, P_{ki}^+)$$

$$\begin{aligned}
r_{ki} &= y_k - h_k(\hat{x}_{ki}^-, 0) \\
S_{ki} &= H_{ki} P_{ki}^- H_{ki}^T + R_k \\
\beta_{ki} &= \frac{\exp\left[-r_{ki}^T S_{ki}^{-1} r_{ki}/2\right]}{(2\pi)^{n/2} |S_{ki}|^{1/2}} \\
a_{ki} &= \frac{a_{k-1,i}\beta_{ki}}{\sum_{j=1}^{M} a_{k-1,j}\beta_{kj}}
\end{aligned}$$